BMAS 1103: ENGINEERING MATHEMATICS III

Course Objectives: To make the students understand the concepts of Statistics, Laplace transform and Complex analysis by giving more emphasis to their applications in engineering.

Credits: 04 **Semester III** **L–T–P: 3–1–0**

Module No.	Contents	Teaching Hours (Approx.)
I	**Statistics:** Introduction, Fitting of straight line by the method of least squares, Correlation, Regression lines, Binomial and Poisson distributions, Sampling, Level of significance, Chi-square test. **Laplace Transform**: Properties of Laplace transform, Laplace transform of derivatives and integrals, Laplace transform of periodic functions, Properties of inverse Laplace transform, convolution theorem, Engineering applications in solving ordinary differential equations.	20
II	**Complex Analysis**: Analytic functions, Harmonic functions, Harmonic conjugates, Milne's Thomson method, Conformal mapping, Bilinear transformation, Line integral in a complex plane, Cauchy's integral theorem, Cauchy's integral formula, Taylor and Laurent series (without proof), Singularities, Residue at a pole, Cauchy's residue theorem and its application in evaluation of real integrals (excluding poles on real axis).	20

Course Outcomes:

After studying these topics, the student will be able to

- **CO 1:** Compute the coefficient of correlation and regression between two variables
- **CO 2:** Apply Chi-square test
- **CO 3:** Use Laplace transform in solving differential equations
- **CO 4:** Know the concept of analytic functions and its applications in engineering
- **CO 5:** Understand bilinear transformation and its applications
- **CO 6:** Use Cauchy's residue theorem in solving real integrals

Text Books:

- M. Goyal and N. P. Bali, A Text Book of Engineering Mathematics, Laxmi Publication, Delhi, 2014.
- B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi, 2015.

Reference Books:

- R. V. Churchill and J. W. Brown, Complex Variables and Applications, McGraw-Hill, New York, 2013.
- S. C. Gupta and V. K. Kapoor, Fundamentals of Mathematical Statistics, Sultan Chand Publications, Delhi, 2014.
- M. K. Jain, S. R. K. Iyengar and R. K. Jain, Advanced Engineering Mathematics, Narosa Publishing House, New Delhi, 2002.

Course Name: **Engineering Mathematics-III**

Course Outcome:

CO 1: Apply method of least square for fitting of straight line

CO 2: Apply Chi-square test

CO 3: Use Laplace transform in solving differential equations

CO 4: Know the concept of analytic functions and its applications in engineering

CO 5: Understand bilinear transformation and its applications

CO 6: Use Cauchy's residue theorem in solving real integrals

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End Term Examination, Odd Semester 2022-23
Program- B. Tech. (CSE, CE, EE, ECE, EN) - II Year, III Semester
Subject Name: Engineering Mathematics-III
Subject Code: BMAS 1103

Time: 3 Hours **Maximum Marks: 50**

Note -- Attempt ALL the sections.

Section – A

*Attempt **All** Questions.* 4 × 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Fit a straight line to the following data: x: 0 1 2 3 4 y: 1 1.8 3.3 4.5 6.3 by the method of least squares.	4	1	A	P
2	Solve the differential equation by using Laplace transform: $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = e^{-t}\sin t,$ with $x(0) = 0,\ x'(0) = 1.$	4	3	U	C
3	A die is thrown 270 times and the results of these throws are given below: No. appeared on the die: 1 2 3 4 5 6 Frequency: 40 32 29 59 57 59 Test whether the die is biased or not. [Given that the tabulated value of χ^2 at 5 % level of significance for 5 d. f. is 11.070.]	4	2	An	F

4	Show that the function defined by $f(z) = \sqrt{\lvert xy \rvert}$ satisfies Cauchy-Riemann equation at the origin but is not analytic at that point.	4	4	E	C
5	Prove that an analytic function with constant modulus is constant. **OR,** Prove that if $f(z) = u + iv$ is an analytic function then the family of curves $u(x, y) = a$ and $v(x, y) = b$ (where a & b are constants) cut each other orthogonally.	4	4	F	C

Section – B

*Attempt **All** Questions* $3 \times 5 = 15$ Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Construct an analytic function $f(z) = u + iv$ in terms of z whose real part is $u = e^x(x\cos y - y\sin y)$.	3	4	A	P
7	Evaluate $\int_C \frac{e^{2z}}{(z+1)^4} dz,$ where C is a circle $\lvert z \rvert = 3$.	3	4	E	C
8	Find the Laurent's series for the function $f(z) = \frac{1}{(z-1)(z-2)}$ in the following regions: (i) $1 < \lvert z \rvert < 2$ (ii) $\lvert z \rvert > 2$.	3	4	A	M

9	Show that the transformation $w = \frac{2z+3}{z-4}$ maps the circle $z\bar{z} - 2(z + \bar{z}) = 0$ into a straight line given by $2(w + \bar{w}) + 3 = 0.$ **OR,** Discuss the following types of singularities of a function f (z) with suitable examples: (a) Removable singularity (b) Essential Singularity (c) Pole	3	5	U	C
10	Find the residue of $f(z) = \frac{z^3}{(z-1)^4(z-2)(z-3)}$ at the singularities which lie inside the circle $\|z - 3\| = 1.5$.	3	4	E	P

Section – C

Attempt All Questions.

$5 \times 3 = 15$ Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Evaluate $\int_0^{2+i} z^2\, dz$ along: (a) the straight line joining $z = 0$ and $z = 2 + i$. (b) the real axis from $z = 0$ to $z = 2$ and then along a line parallel to imaginary axis from $z = 2$ to $z = 2 + i$. (c) the imaginary axis from $z = 0$ to $z = i$ and then along a line parallel to real axis from $z = i$ to $z = 2 + i$.	5	4	U	P

12	Find the bilinear transformation which maps the points $z = -1, 0, 1$ of the z-plane onto the points $w = -1, -i, 1$ of the w-plane. Show that under this transformation, the upper half of the z-plane maps onto the interior of the unit circle $\|w\| = 1$. **OR,** Define harmonic function. Show that $u = \frac{1}{2} log_e(x^2 + y^2)$ is a harmonic function. Also find its harmonic conjugate v such that $u + iv = f(z)$ is an analytic function of $z = x + iy$. Hence, determine the function $f(z)$ in terms of z.	5	5	A	P
13	Evaluate by the method of contour integration: $\int_0^{2\pi} \frac{1}{(2 + \cos\theta)} d\theta.$ **OR,** Determine the poles of the function $f(z) = \frac{z^2}{(z-1)^2 (z+2)}$ and the residues of $f(z)$ at each of its poles. Hence evaluate $\int_c f(z)\, dz$ where $c \equiv \|z\| = 3$ using Cauchy residue theorem.	5	6	An	C

Course Outcomes:

CO 1: Compute the coefficient of correlation and regression between two variables, Binomial and Poisson, distributions, Sampling, Level of significance, Chi-square test.

CO 2: Use Laplace transform in solving differential equations

CO 3: Know the concept of analytic functions and its applications in engineering

CO 4: Understand bilinear transformation and its applications

CO 5: Use Cauchy's residue theorem in solving real integrals

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End Term Examination, Even Semester 2021-22
B. Tech. (CS, ME), II Year, IV Semester
Subject Code & Subject Name- BMAS 1103, ENGINEERING MATHEMATICS III

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

(1) Attempt all the sections.

(2) Marks of the questions and internal choice are indicated in each section.

Section – A

*Attempt **All** Questions.* 4 X 5 = 20 Marks

<table>
<tr><th>No.</th><th>Detail of Question</th><th>Marks</th><th>CO</th><th>BL</th><th>KL</th></tr>
<tr><td>1</td><td>Find the Laplace transform of
$f(t) = \cosh(2t)\sin(4t)$
and hence, evaluate the integral
$\int_0^\infty e^{-10t}\cosh(2t)\sin(4t)dt$</td><td>4</td><td>2</td><td>E</td><td>C</td></tr>
<tr><td>2</td><td>Solve the following initial value problem by the help of Laplace transform:
$(D^2 - 2D + 2)\,y = 0,\ y = Dy = 1 \text{ at } t = 0.$</td><td>4</td><td>2</td><td>A</td><td>P</td></tr>
<tr><td>3</td><td>A skilled typist, on routine work, kept a record of his typing mistakes made per day for 300 days, in following format:
<table>
<tr><td>Mistakes per day</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr>
<tr><td>No. of days</td><td>143</td><td>90</td><td>42</td><td>12</td><td>9</td><td>3</td><td>1</td></tr>
</table>
Fit a Poisson distribution to the above data and hence, calculate the theoretical frequencies.</td><td>4</td><td>1</td><td>An</td><td>P</td></tr>
</table>

4	Prove that the following function is harmonic. $u = x^2 - y^2 - 2xy - 2x + 3y$ Also find the harmonic conjugate of this function.	4	3	U	C
5	Find the residue of the following function at simple poles: $f(z) = \frac{2z^2 + 5}{(z+2)^3(z^2+4)}$, and hence, evaluate the complex line integral $\oint_c f(z)dz$, where, c is the boundary of the square formed with the vertices $1+i, 2+i, 2+2i, 1+2i$. **OR,** The following table gives a classification of a sample of 160 plants of their flower color and flatness of leaves.	4	4	E	F

	Flat leaves	Colored leaves	Total
White flower	99	36	135
Red flower	20	5	25
Total	119	41	160

Test if the flower color is independent of the flatness of leaves.
[Given: The value of χ^2 at 5 % level of significance for 1 degree of freedom is 3.841.]

1

Section – B

*Attempt **All** Questions.* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Using Cauchy-Riemann theorem, show that $f(z) = z^3$ is analytic on the whole complex plane.	3	3	R	F
2	Using Milne-Thomson method, find the analytic function $f(z) = u + iv$ if it is given that $v = e^x(x\sin y + y\cos y)$.	3	3	An	C

3	Evaluate the contour integral using Cauchy's residue theorem: $\int_c \frac{2z+1}{z^2+z} dz$, where, c is the circle given by $\mid z \mid = \frac{1}{2}$.	3	5	A	P
4	Expand the following function in Laurent's series valid for $1 < \mid z \mid < 3$: $f(z) = \frac{1}{(z+1)(z+3)}$	3	5	C	M
5	Find the residue of the following function at each of its poles: $f(z) = \frac{1}{(z-1)(z-2)(z-3)}$	3	5	E	P

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Evaluate the integral $\int_0^{\pi} \frac{d\theta}{2\cos\theta + 3}$ by method of contour integrals in complex plane.	5	5	A	C
2	Examine the nature of the function $f(z) = \frac{x^3 y(y-ix)}{x^6 + y^2}; z \neq 0$ and $f(0) = 0$. Show that $\lim_{z \to 0} \frac{f(z) - f(0)}{z} \to 0$ along radius vector $y = mx$ for any value of m. Is $f(z)$ analytic? Support your answer with proof.	5	3	C	M
3	Evaluate the integral: $\int_0^{1+i} (x - y + i x^2)\, dz$ along (a) the straight line joining z = 0 and z = 1 + i (b) the real axis from z = 0 to z = 1 and then along a line parallel to imaginary axis from z = 1 to z = 1 + i.	5	3	An	P

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-

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End Term Examination, Odd Semester 2021-22
B. Tech., CS (Sec. E, F, G, H, I, J, K, L), EE, EN, EC and CE,
II Year, III Semester
BMAS 1103: Engineering Mathematics III

Time: 3 Hours **Maximum Marks: 50**

Section- A

Note: Attempt ANY FIVE Questions. 5 x 4 = 20 marks

Q.1. Comment on the following statements:

(a) For a binomial distribution, mean is 6 and variance is 9.

(b) For a Poisson distribution with number of trials n = 1000 and probability of happening of an event p = 0.002, mean and variance are 2.

Q.2. The regression lines of y on x and x on y are respectively,

$y = a x + b,$

$x = c y + d.$ (a, b, c and d are constants)

Can you determine the mean values of x and y? If yes, find them.

Q.3. Evaluate: $L^{-1}[\frac{p+8}{p^2+4p+5}]$

Q.4. What are Cauchy- Riemann equations in polar coordinates? Show that $f(z) = r^2(\cos 2\theta + i \sin 2\theta)$ is analytic.

Q.5. What is Cauchy's integral theorem? Evaluate:

$$\oint_c \frac{e^{-z}}{z+1} dz$$

where c is the circle with centre at z = 0 and radius 0.5.

Q.6. What is a singularity? Discuss the nature of singularity of

$$f(z) = \frac{z - \sin z}{z^3} \quad \text{at } z = 0.$$

Section- B

Note: Attempt ALL Questions. 5 x 3 = 15 marks

Q.1. Examine the nature of the function

$$f(z) = \frac{x^2y^5(x+iy)}{x^4+y^{10}} ; z \neq 0 \text{ and } f(0) = 0$$

in the region including the origin.

Q.2. Determine the constants a, b, c and d so that the function

$$f(z) = (x^2 + a\,x\,y + by^2) + i\,(c\,x^2 + d\,x\,y + y^2)$$

is analytic.

Q.3. Expand the function $f(z) = \frac{1}{z-2} - \frac{1}{z-1}$ in the regions:

(a) $|z| < 1$

(b) $1 < |z| < 2$

(c) $|z| > 2$.

Q. 4. Show that:

$$\oint_C \frac{e^z}{z}\, dz = 2\pi i \quad \text{where c is } |z| = 1.$$

Hence show that,

$$\int_0^{2\pi} e^{\cos\theta} \cos(\sin\theta)\, d\theta = 2\pi$$

Q. 5. Find the pole of the function

$$f(z) = \frac{1 - e^{2z}}{z^4}$$ and determine its order.

Find also the residue of $f(z)$ at that pole.

Section- C

Note: Attempt ANY THREE Questions. 3 x 5 = 15 marks

Q.1. Using contour integration, evaluate:

$$\int_0^{2\pi} \frac{d\theta}{a + b\cos\theta} \; ; a > |b|$$

Q.2. If $u - v = (x - y)(x^2 + 4xy + y^2)$ and $f(z) = u + iv$ is an analytic function of $z = x + iy$ then find $f(z)$ in terms of z.

Q.3. What is the bilinear transformation which maps the points $z = \infty, i, 0$ of z – plane into the points $w = 0, i, \infty$ of w- plane?

Q.4. Integrate $f(z) = Re\,(z)$ from $z = 0$ to $z = 1 + 2i$,

(a) Along the straight line joining $z = 0$ and $z = 1 + 2i$.

(b) Along the real axis from $z = 0$ to $z = 1$ and then along a line parallel to imaginary axis from $z = 1$ to $z = 1 + 2i$.

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End Term Examination, Odd Semester 2019-20

B. Tech., (Common to all Branches), II-Year, III-Semester

BMAS 0103 : Engineering Mathematics-III

Time: 3 Hours **Max Marks: 50**

Section-A

Note: Answer ALL the questions. **(7x5=35)**

I. Obtain the half range sine series of the function

$$f(x)=kx(x-l)\ in\quad 0\le x\le l.$$

II. Solve the partial differential equation by the method of separation of variables:

$$3\frac{\partial u}{\partial x}+2\frac{\partial u}{\partial y}=0,\quad u(x,0)=4e^{-x}.$$

(OR)

Derive the various possible solutions of one-dimensional heat equation

$$\frac{\partial u}{\partial t}=c^2\frac{\partial^2 u}{\partial x^2}$$

by the method of separation of variables.

III. Use Laplace transform to evaluate:

$$\int_0^\infty te^{-3t}\cos 2t\ dt$$

(OR)

Find the Laplace transform of the half sine wave rectifier function

$$F(t) = \begin{cases} E \sin wt, & 0 < t < \frac{\pi}{w} \\ 0, & \frac{\pi}{w} < t < \frac{2\pi}{w} \end{cases}$$

and $F\left(t+\frac{2\pi}{w}\right) = F(t)$.

IV. Prove that an analytic function with constant modulus is constant.

V. Expand the following function in Laurent's series:

$$f(z) = \frac{z}{(z^2+1)(z^2+4)} \quad in \quad 1<|z|<2$$

VI. Evaluate the following integral using contour integration:

$$\int_0^{2\pi} \frac{d\theta}{5+3\cos\theta}$$

VII. Find the inverse Laplace transform of the following function:

$$f(p) = \frac{p^2}{(p^2+4)(p^2+9)}$$

(OR)

What is Fourier series? What must be the conditions that the function must possess for its Fourier series expansion? Also, expand the following function in Fourier series:

$$F(x) = \begin{cases} x, & 0 < x < 1 \\ 0, & 1 < x < 2 \end{cases}$$

Section-B

Note: Answer ALL the questions. **(3x2=6)**

I. Prove that $u = \frac{1}{2}\log\left(x^2 + y^2\right)$ is harmonic.

II. 2. Expand $f(z) = \frac{1}{z}$ as a Taylor's series about the point $z = 1$.

III. 3. Calculate the residue of $F(z) = \frac{z^2}{(z-1)(z-2)^2}$ at each of its singularity and hence find

$$\oint_c F(z)dz \qquad \text{where, c is } |z| = 1.5$$

Note: Answer ALL the questions. **(3x3=9)**

I. Prove that $f(z) = z^n$ is analytic; n is a constant.

II. State and prove Cauchy's integral theorem.

III. Find the residue of $f(z) = \frac{z\sin z}{(z-\pi)^3}$.

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END Term Examination, 2018–2019

Sub.:- Engineering Mathematics III (BMAS–0103)

Time: - 3 Hrs. **Course:- B. Tech. III Sem.** **Max. Marks:- 50**

Note - Attempt **BOTH** Sections.

SECTION A (7 x 5 = 35 marks)

Note: ALL questions of this section are **COMPULSORY.** Each question is of **Five** marks.

Q.1. Obtain the Fourier series to represent $F(x) = \frac{1}{4}(\pi - x)^2$ in the interval $0 \leq x \leq 2\pi$.
Hence or otherwise evaluate:

(a) $\sum_{k=1}^{\infty} \frac{1}{k^2}$ (b) $\sum_{s=1}^{\infty} \frac{1}{(2s-1)^2}$

Q.2. Find the half range sine series of $F(x) = lx - x^2$ in the interval $(0, l)$. Hence deduce that,

$$\frac{\pi^3}{32} = 1 - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots$$

Q.3. An infinitely long plane uniform plate is bounded by two parallel edges and an end at right angles to them. The breadth is π. This end is maintained at temperature u_0 at all points and the other edges are at zero temperature. Determine the temperature at any point of the plate in the steady state.

Q.4. Given that: $L(t \sin \omega t) = \frac{2\omega p}{(p^2+\omega^2)^2}$ (ω is a constant).

Using it, evaluate:

(a) $L(\omega t \cos \omega t + \sin \omega t)$

(b) $L(3 \sin \omega t + \omega t \cos \omega t)$

Q.5. Use convolution theorem to evaluate:

$$L^{-1}\left[\frac{1}{p(p+1)(p+2)}\right]$$

Q.6. If f (z) = u + iv is an analytic function of z = x+ iy, find f (z) in terms of z if

$$2u - v = e^x(2\cos y - \sin y).$$

Q.7. Using contour integration, evaluate: $\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta}; a > |b|$

Hence or otherwise evaluate: $\int_0^{2\pi} \frac{d\theta}{\sqrt{2}-\cos\theta}$

SECTION B (15 marks)

Note: Attempt **ALL** questions. Marks are shown against them.

Q.1. An electrostatic field in the xy-plane is given by potential function $\varphi\,(x, y) = 3x^2y - y^3$, find stream function $\psi\,(x, y)$ if $\varphi + i\,\psi$ is an analytic function of z. (2)

Q.2. Write a short note on singularities of a complex function and its types giving suitable examples. (2)

Q.3. Using Cauchy's residue theorem, evaluate: (3)

$$\int_c \frac{3z^2+z+1}{(z-1)(z+1)(z+3)}\,dz$$

where c is the circle $|z| = 4$.

Q.4. State Cauchy's integral formula.
Hence or otherwise evaluate:

$$\int_c \frac{e^{i\pi z}}{2z^2-5z+2}\,dz$$

where c is a unit circle with centre at origin and having positive orientation. (1 + 2)

Q.5. Evaluate $\int_0^{1+i}(x-y+ix^2)\,dz$ along, (3)

(a) the straight line y = x.

(b) the real axis from z = 0 to z = 1 and then along a line parallel to imaginary axis from z = 1 to z = 1+ i.

Q.6. Expand $f(z) = \frac{1}{z^2-3z+2}$ in the following regions: (2)

(a) $|z| < 1$ (b) $0 < |z-1| < 1$

End Term Examination, Odd Semester, 2017-18

B.Tech. (II Year)-III Semester

Subject:- Engineering Mathematics –III **Subject Code:- AHM-3101**

Time : 02 1/2 Hours Max. Marks: 40

Section- A

Note: Attempt All Questions. 1 x 16 = 16 marks

(I) Find the inverse Laplace transform of $\frac{1}{(s+a)^2}$.

(II) Find the Laplace transform of $\delta(t-a)$.

(III) Find the Laplace transform of $\sin^2 3t$.

(IV) Find the Laplace transform of 4^t.

(V) Apply convolution theorem to evaluate $t * t$.

(VI) Write the condition(s) under which the Laplace transform of the derivative of F (t) exists.

(VII) Evaluate $\int_c \frac{\cos \pi z}{z-1} dz$, if c is $|z| = \frac{3}{2}$.

(VIII) For the function $f(z) = \frac{1}{z(e^z - 1)}$, find the order of the pole at z= 0.

(IX) Evaluate $\int_0^{1+i} (x^2 - iy)\,dz$, along the path y = x.

(X) Find the residue of $f(z) = \frac{z^2 - 1}{(z-1)^2(z+2)}$, at the simple pole.

(XI) A rod 50 cm long with insulated sides has its ends A and B kept at $20^\circ c$ and $70^\circ c$ respectively.

Find the steady state temperature distribution of the rod.

(XII) Solve $\frac{\partial^2 z}{\partial x^2}+2\frac{\partial^2 z}{\partial x \partial y}+\frac{\partial^2 z}{\partial y^2}=0$.

(XIII) Classify the following partial differential equation $x^2\frac{\partial^2 z}{\partial x^2}+\left(1-y^2\right)\frac{\partial^2 z}{\partial y^2}=0$, where $-\infty < x < \infty, x \neq 0, -1 < y < 1$.

(XIV) Solve $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=z$.

(XV) Find the particular integral of the partial diff. equation $\frac{\partial^2 z}{\partial x^2}-2\frac{\partial^2 z}{\partial x \partial y}=e^{2x}$.

(XVI) Solve $\left(\frac{\partial z}{\partial x}+4\frac{\partial z}{\partial y}+5\right)^2=0$

Section- B

Note: Attempt Any Four Questions. **3 x 4 = 12 marks**

Q1 If $f\left(x+iy\right)=x^3-3xy^2+i\phi(x,y)$, where $i=\sqrt{-1}$ and $f(x+iy)$ is an analytic function then find $\phi(x,y)$.

Q2 Using Cauchy's integral theorem, find the value of the integral (integral being taken in contour clockwise direction) $\oint_c \frac{z^3-6}{3z-i}dz$, where c is $|z|=1$ and z = x+ i y.

Q3. If $f(z)=c_0+c_1z^{-1}$, then evaluate $\oint_{|z|=1}\frac{1+f(z)}{z}dz$.

Q4. Given $\phi(z)=\frac{z}{(z-a)^2}$ with $|z|>a$, find the residue of $\phi(z)z^{n-1}$ at z = a, for $n \geq 0$.

Q5. Find the analytic region of $f(z)=(x-y)^2+2i(x+y)$.

Section- C

Note: Attempt Any Three Questions. **4 x 3 = 12 marks**

Q1. Show that the function $u(x,y)=3x^2y+2x^2-y^3-2y^2$ is harmonic. Find the conjugate harmonic function $v(x,y)$.

Q2. Expand $\frac{1}{z^2-3z+2}$ in the region $(i)1<|z|<2,\ (ii)0<|z-1|<1$.

Q3. Evaluated by Cauchy's residue theorem, $\int_c \frac{z-1}{(z+1)^2(z-2)}dz$ where c is $|z-i|=2$.

Q4. Evaluate $\int_0^{2\pi} \frac{d\theta}{13+5\sin\theta}$.

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End Term Examination, 2016 – 17

B. Tech. II Year III Sem. (Common to ALL branches)

Subject: - ENGINEERING MATHEMATICS - III (AHM – 3101)

Time: - 2 Hrs. 30 Minutes **Total Marks:- 40**

Note: Attempt **ALL** sections. The terms, used in question paper, have their usual meanings.

SECTION A (16 x 1 = 16 marks)

Instructions: This section has **16** questions. Attempt **ALL** questions of this section.

Q.1. Mention any two applications of partial differential equations in Engineering.

Q.2. Solve the partial differential equation: $\frac{p}{\cos x}+\frac{q}{\cos y}=\sec z$

Q.3. Find the complementary function (C. F.) of the partial differential equation:

$$\frac{\partial^4 z}{\partial x^4}-\frac{\partial^4 z}{\partial y^4}=0$$

Q.4. Find the particular integral (P.I.) of the partial differential equation:

$$4r - 4s + t = 16 \log (2y + x).$$

Q.5. Solve the partial differential equation: $r + 2s + t = 0$

Q.6. Classify the partial differential equation: $u_{xx} + u_{yy} = u$.

Q.7. Find the Laplace transform of

$$F(t) = 2017$$

Q.8. Find the inverse Laplace transform of

$$f(p) = \frac{2p+6}{p^2+4}$$

Q.9. Find the Laplace transform of

$$F(t) = t e^t$$

Q.10. Find the inverse Lapiace transform of

$$f(p) = \frac{p}{p^2+1} e^{-\frac{\pi}{3}p}$$

Q.11. Find the Laplace transform of

$$F(t) = \frac{\sin 2t}{t}$$

Q.12. Find the inverse Laplace transform of $f(p) = \frac{1}{p^2(p^2+1)}$

Q.13. Evaluate $\oint_c \frac{dz}{z-2}$ around the circle $|z-2|=4$.

Q.14. What are Cauchy – Riemann equations?

Q.15. Show that the function $v = 3x^2y - y^3$ is harmonic.

Q.16. State Cauchy integral theorem.

SECTION B (4 x 3 = 12 marks)

Instructions: This section has 5 questions. Attempt **ANY FOUR** questions of this section.

Q.1. If f (z) is analytic within a circle C given by $|z-a|=R$ and if $|f(z)|<M$ on C, then prove that,

$$|f^n(a)| \leq \frac{M\lfloor n}{R^n}$$

Q.2. Evaluate: $\oint_c \frac{z^2+1}{z^2-1}dz$

where c is the circle:

(i) $|z|=3$

(ii) $|z-1|=1$

Q.3. Evaluate: $\oint_c \frac{1}{z^2+9}dz$ where c is $|z-3i|=4$

Q.4. (i) Discuss the nature of singularity of f (z) = $\frac{\sin(z-1)}{z-1}$ at z =1

(ii) Prove that f (z) = $\frac{e^z+e^{-z}}{2}$ is analytic.

Q.5. Evaluate $\int_0^{3+i} z^2 dz$ along the straight line $y=\frac{x}{3}$.

SECTION C (3 x 4 = 12 ma

Instructions: This section has 4 questions. Attempt **ANY THREE** questions of this sec

Q.1. Find the poles of the function $f(z) = \frac{z}{(z-1)(z-2)}$ and residues of f (z) at each of its poles. Also, evaluate $\oint_c f(z)dz$ using residue theorem where C is the circle $|z| = 3$.

Q.2. An electrostatic field in the xy – plane is given by the potential function $\Phi = x^2 - y^2$. Find the stream function Ψ if $f(z) = \Phi + i\,\Psi$ is analytic. Also find f (z) in terms of z.

Q.3. Expand the function $f(z) = \frac{1}{(z+1)(z-3)}$ in the following regions:

(i) $|z-1| < 2$

(ii) $-1 < |z| < 4$

(iii) $|z| > 4$

Q.4. Using contour integration, prove that:

$$\int_0^\infty \frac{\log(1+x^2)}{1+x^2}dx = \pi \ln 2$$

End Term ODD SEMESTER Examination, 2015 – 16

B.Tech. II Year III Sem. (Common to ALL branches)

Subject: - ENGINEERING MATHEMATICS - III (AHM – 3001)

Time: - 2 Hrs. 30 Minutes **Total Marks:- 40**

Note: Attempt **ALL** sections. The terms, used in question paper, have their usual meanings.

SECTION A

(16 x 1 = 16 marks)

Instructions: This section has **16** questions. Attempt **ALL** questions of this section.

Q.1. Form the partial differential equation from the relation :

$$z = a x + by + a b$$

Q.2. Solve the partial differential equation :

$$p x + q y = z .$$

Q.3. Solve the partial differential equation :

$$r - 4s + 4t = 0.$$

Q.4. Find the particular integral of the partial differential equation:

$$r + 2s + t = 2 (y - x).$$

Q.5. Solve the partial differential equation :

$$r + 2s + t + 2p + 2q + z = 0$$

Q.6. Classify the partial differential equation :

$$u_{xx} + u_{yy} + u_{zz} = 0.$$

Q.7. Find the Laplace transform of $F(t) = 1^t$

Q.8. Find the Laplace transform of

$$F(t) = \lfloor 2$$

Q.9. Find the Laplace transform of

$$F(t) = t \sin t$$

Q.10. Find the inverse Laplace transform of

$$f(p) = \frac{\sqrt{2016}}{p^{2016}}$$

Q.11. Find the inverse Laplace transform of

$$f(p) = \frac{1}{p^2 + 2p + 2}$$

Q.12. Find the inverse Laplace transform of

$$f(p) = \cot^{-1} p$$

Q.13. Prove that $u = e^x \cos y$ is a harmonic function.

Q.14. What are Cauchy – Riemann equations in Polar Coordinates?

Q.15. Prove that :

$$\oint_c \frac{dz}{z-a} = 2\pi i \qquad \text{where c is a circle } |z-a| = r$$

Q.16. What is the nature of singularity of the function

$$f(z) = \sin\left(\frac{1}{z-a}\right) \text{ at } z = a \text{ ?}$$

SECTION B

(4 x 3 = 12 marks)

Instructions: This section has 5 questions. Attempt **ANY FOUR** questions of this section.

Q.1. If f (z) = u + iv is an analytic function, prove that u and v will be harmonic functions.

Q.2. If $v = \tan^{-1}(y / x)$ is the imaginary part of an analytic function f(z) = u + iv then find f (z) in terms of z.

Q.3. Evaluate: $\oint_c \frac{z+4}{z^2+2z+5}dz$ where c is

(i) $|z-1|=1$

(ii) $|z-1-i|=2$.

Q.4. Prove that an analytic function w = u + iv cannot be expressed as a function of $\bar{z}$.

Q.5. Evaluate using contour integration:

$$\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} \quad ; \quad a > |b| > 0$$

SECTION C

(3 x 4 = 12 marks)

Instructions: This section has 4 questions. Attempt **ANY THREE** questions of this section.

Q.1. If f (z) = u + iv is a regular function of z, prove that :

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2$$

Q.2. Evaluate $\int_a^{1+i}(x-y+ix^2)dz$ along:

(i) the straight line y = x

(ii) the real axis from z = 0 to z = 1 and then along a line x = 1 from z = 1 to z = 1+ i.

Q.3. Expand $f(z) = \frac{1}{z^2-3z+2}$ in the Taylor or Laurent's series in the following regions:

(i) $|z| < 1$

(ii) $|z| > 2$

(iii) $0 < |z-1| < 1$

Q.4. Find the residue of

$$f(z) = \frac{z^3}{(z-1)^4(z-2)(z-3)}$$ at its poles.

Hence evaluate $\oint_c f(z)dz$, using Cauchy's residue theorem, where c is the circle $|z| = 4$

End Term ODD SEMESTER Examination, 2014 - 15

B.Tech. II Year III Sem. (Common to ALL branches)

Subject: - MATHEMATICS - III (AHM – 201)

Time:- 2 Hrs. 30 Minutes **Total Marks:- 40**

Note:

(1) This question paper contains three sections – A, B and C. Section A contains 16 questions, Section B contains 5 questions while Section C contains 4 questions.

(2) Attempt **ALL** questions from Section A, **ANY FOUR** questions from Section B and **ANY THREE** questions from Section C.

(3) Any missing data may be assumed suitably giving proper justification.

SECTION A (16 x 1 = 16)

Instructions: This section has **16** questions. Attempt **ALL** questions of this section.

Q.1. Define Fourier transform of a function F(x).

Q.2. State and Prove linear property of Fourier transform.

Q.3. Solve the integral equation: $\int_0^\infty F(x)\cos px\,dx = \begin{cases} 1-p, 0 \le p \le 1 \\ 0, p > 1 \end{cases}$

Q.4. Find the Z – transform of $\{f(k)\}$ where $f(k) = k\,(k-1)$ where $k \geq 0$

Q.5. Given that $Z\{u_k\} = \frac{2z^2+3z+4}{(z-3)^2}$, $|z| > 3$. Show that $u_1 = 7$.

Q.6. Find the Z – transform of unit step sequence u (k).

Q.7. Let u(x,y) be the real part of an entire function $f(z) = u(x,y) + i\,v(x,y)$ for $z = x + iy \in C$. If C is the positively oriented boundary of a rectangular region R in R^2 then evaluate $\oint_c \left(\frac{\partial u}{\partial y}dx - \frac{\partial u}{\partial x}dy\right)$

Q.8. If $\vec{\phi}$ and $\vec{\psi}$ are irrotational vectors then prove that

$$\text{div}\,(\vec{\phi} \times \vec{\psi}) = 0$$

Q.9. Find the directional derivative of $f(x, y, z) = x + y + z$ at the point $(\frac{3}{4}, \frac{1}{9}, \frac{5}{7})$ in the direction parallel to y – axis.

Q.10. Prove that: $\{\text{grad } f(r)\} \times \vec{r} = \vec{0}$ where $\vec{r} = x\,i + y\,j + z\,k$.

Q.11. Define analytic function.

Q.12. If $u = x + y + z$, $v = x^2 + y^2 + z^2$ and $w = xy + yz + zx$ then show that grad u, grad v and grad w are coplanar vectors.

Q.13. What are Cauchy – Riemann equations?

Q.14. Prove that $u = x^2 - y^2$ is a harmonic function.

Q.15. Define singularity of a function f (z) with a suitable example.

Q.16. Show that: curl (grad φ) = $\vec{0}$ where φ is a scalar point function.

SECTION B

(4 x 3 = 12)

Instructions: This section has 5 questions. Attempt **ANY FOUR** questions of this section.

Q.1. Find the harmonic conjugate of $u = x^3 - 3xy^2$. Also construct the corresponding analytic function $f(z) = u + iv$ and express it in terms of z.

Q.2. Let $f(z) = u(x, y) + i\,v(x, y)$ be an analytic function of $z = x + i\,y$. Then using calculus of residues, Show that:

$$\int_0^{2\pi} f(e^{i\theta}) \cos\theta \, d\theta = \pi f'(0)$$

Q.3. Expand the function

$$f(z) = \frac{1}{z} - \frac{3}{z+1} + \frac{2}{z-2}$$

in the region $0 < |z+1| < 1$

Q.4. Evaluate: $\oint_c \frac{(z^2+1)}{(z^2-1)} dz$ where c: $|z| = \frac{3}{4}$

Q.5. Let C be the positively oriented circle given by $|z+3i| = 2$ then evaluate:

$$\oint_c \frac{dz}{z^2+9}$$

SECTION C

(3 x 4 = 12)

Instructions: This section has 4 questions. Attempt **ANY THREE** questions of this section.

Q.1. If $f(z) = u + iv$ is an analytic function of $z = x + iy$, find $f(z)$ in terms of z if

$$\alpha u - \beta v = e^x (\alpha \cos y - \beta \sin y), \alpha \neq 0, \beta \neq 0$$

Q.2. Expand the function $f(z) = \frac{1}{(z+1)(z+3)}$ in the following regions:

(a) $1 < |z| < 3$

(b) $|z| > 3$

Q.3. Determine the poles of the function

$$f(z) = \frac{z^2}{(z-2)^2(z+3)}$$

and the residue at each pole. Hence evaluate $\int_c f(z)dz$ where c is a circle, with centre at origin and of radius 4.

Q.4. Using contour integration, Prove that:

$$\int_0^\infty \frac{dx}{(1+x^2)^2} = \frac{\pi}{4}$$

Uni. Roll No:- _____

GLA University, Mathura

End Term ODD SEMESTER Examination, 2013 - 14

Course: - B.Tech. II Year III Sem. (Common to ALL branches)

Subject: - MATHEMATICS - III (AHM – 201) Time:- 2 Hrs. 30 Minutes Total Marks:- 40

Note: (1) This question paper has two pages. It contains three groups – A, B and C. Group A contains 10 questions, Group B contains 6 questions while group C also contains 6 questions.
(2) Attempt **ANY EIGHT** questions from Group A, **ANY FOUR** questions from Group B and **ANY FOUR** questions from Group C.
(3) Any missing data may be assumed suitably giving proper justification.
(4) The instructions provided for a particular group should strictly be followed. Do not attempt extra questions of a particular group otherwise the first solved eight questions of group A, first solved four questions of group B & first solved four questions of group C will be evaluated only.

GROUP A (8 x 1 = 8)

Instructions: This group has 10 questions. Attempt **ANY EIGHT** questions of this group.

Q.1. Define gradient of a scalar point function $\varphi(x, y, z)$. Also find grad f (r) where $r = |\vec{r}|$ and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

Q.2. Obtain a new biquadratic equation after multiplying the roots of $x^4 + 2x^3 + 4x^2 + 6x + 8 = 0$ by 1/2.

Q.3. Using Green's theorem, evaluate $\int_c (xydy - y^2dx)$ where c is the square cut from the first quadrant by the lines x =1, y =1.

Q.4. Find the inverse Z – transform of $F(z) = \frac{z}{(z-1)^4}$, $|z| > 1$

Q.5. Solve the integral equation for F(x) : $\int_0^\infty F(x)\sin \lambda x dx = (\pi/2)\, e^{-\lambda}$

Q.6. "If f(z) = u + iv is an analytic function of z then u and v both will be harmonic. But the converse of this theorem may not always be true." Explain by taking suitable examples.

Q.7. Find the Fourier transform of $F(x) = e^{-a|x|}$, $a > 0$

Q.8. If we want to have an expansion of a function f(z) about one of its singular points then in which of the power series may the function f(z) be expanded ? Explain.

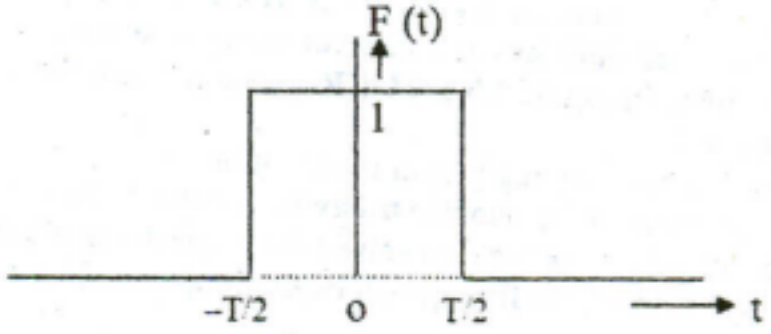
Q.9. Give an example of a function having residue at infinity yet analytic there.

Q.10. Evaluate $\int_c (5z^4 - z^3 + 2)dz$ around a curve consisting of the parabola $y = x^2$ from (0,0) to (1,1) and $y^2 = x$ from (1,1) to (0,0).

GROUP B (4 x 3 = 12)

Instructions: This group has 6 questions. Attempt **ANY FOUR** questions of this group.

Q.1. Show that the Fourier transform of the single gate function (rectangular pulse) shown in the following figure is T sinc $(\frac{pT}{2})$ where sinc $\theta = \frac{\sin\theta}{\theta}$:



Q.2. Let $f(z) = u(r, \theta) + iv(r, \theta)$ be an analytic function. If $u(r, \theta) = -r^3 \sin 3\theta$ then construct the corresponding analytic function f(z) and finally express it in terms of z.

Q.3. Find Taylor's series expansion of $f(z) = \frac{2z^3+1}{z(z+1)}$ about the point $z = 1$.

Q.4. Find the Z – transform of $\{f(k)\}$ where $f(k) = c^k \cos(\alpha k)$, $k \geq 0$ and c is a constant.

Q.5. Solve by Cardon's method: $x^3 - 3x^2 + 3 = 0$

Q.6. Evaluate using Cauchy's residue theorem: $\oint_c z^2 e^{\frac{1}{z}} dz$ where $c \equiv |z| = 1$

GROUP C (4 x 5 = 20)

Instructions: This group has 6 questions. Attempt **ANY FOUR** questions of this group.

Q.1. Find an analytic function $f(z) = u + iv$ in terms of z such that $f(0) = 0$ and whose imaginary part is $e^{-x}(x \cos y + y \sin y)$

Q.2. If $w = f(z)$ is an analytic function of $z = r e^{i\theta}$ then prove that $\frac{dw}{dz} = e^{-i\theta}\frac{\partial w}{\partial r}$

Q.3. Using contour integration, evaluate : $\int_{-\infty}^{\infty} \frac{dx}{(a^2+x^2)(x^2+b^2)}; a > b > 0$

Q.4. Use contour integration to evaluate : $\int_0^{\pi} \frac{\cos 3\theta}{5 - 4\cos\theta} d\theta$

Q.5. If $f(z) = \frac{z^2}{(z-1)^3(z-2)(z-3)}$ then evaluate $\int f(z)dz$ using Cauchy's residue theorem where c is the circle $|z| = 9/4$.

Q.6. Expand $f(z) = \frac{7z^2+9z-18}{z^3-9z}$ in Laurent's series valid for the following regions:

(i) $0 < |z| < 3$ (ii) $|z| > 3$

GLA University, Mathura **A**

End-Term Examination, 2012-13

Course: - B.Tech. **II-Year, III-Sem.**

Subject: - Mathematics-III (AHM-201) **Uni. Roll No:-**

Time:-2Hour 30 Minutes **Total Marks:-40**

Notes:-

1.Attempt **ALL** questions of Section- A, **ANY FOUR** from Section- B, and **ANY FOUR** from Section- C.
2.All parts of a question (a, b, etc.) should be answered at one place.
3.Answer should be brief and to-the-point and be supplemented with neat sketches.
4.Any missing or wrong data may be assumed suitably giving proper justification.

Section - A (1 × 10 = 10)

Q.1 The nature of singularity of $f(z)=\sin\frac{1}{z-1}$ at $z=1$ is

* Removable * Isolated essential * Non Isolated essential * Non of these

Q.2 The value of $\int \frac{dz}{z^2+9}$, where C is $|z-3i|=4$ is

* $\frac{\pi}{3}$ * $-\frac{\pi}{3}$ * 0 * $-\frac{\pi}{4}$

Q.3 If $f(x)=x\sin x$ in $(0,2\pi)$, then value of a_0 is

* 0 $f(x)=x\sin x$ * -1 * 2 * -2

Q.4 The equation whose roots are decrease by 2 to the roots of the eqn. $x^3-6x^2+12x-8=0$ is

* $x^3-x^2-8=0$ * $x^3-8=0$ * $x^3-3x^2+3x-1=0$ * $x^3=0$

Q.5 The function $f(z)=e^z$ has an isolated essentia! singularity at

* 0 * -1 * $\frac{\pi}{3}$ * None of these

Q.6 the Cauchy Reimann equations in polar form are

* $ru_r=v_\theta, rv_r=-u_\theta$ * $u_r=rv_\theta, rv_r=-u_\theta$ * $v_\theta=\frac{1}{r}u_r, u_\theta=-\frac{1}{r}v_r$ * $ru_r=v_\theta, rv_r=u_\theta$

Q.8 The Z- transforms of $\delta(k-n)$ is

* z^n * $z^{1/n}$ * $1/z^n$ * z

Q.8 The Fourier sine transforms of e^{-x} is

* $\frac{p}{p^2-1}$ * $\frac{p}{1-p^2}$ * $\frac{p}{p^2+1}$ * $\frac{1}{p}$

Q.9 Using the fact that $\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}$, $\sum_{n=1}^{\infty}\frac{1}{(2n+1)^2}$ is equal to

* $\frac{\pi^2}{12}$ * $\frac{\pi^2}{12}-1$ * $\frac{\pi^2}{8}$ * $\frac{\pi^2}{8}-1$

Q.10 Let $f:C\to C$ be a complex function of the form $f=u(x,y)+iv(x,y)$ and suppose that $u(x,y)=3x^2y$, then

* f cannot be analytic for any v * f is analytic for suitable v * f is analytic for all v * none of these

Section - B (2.5 × 4 = 10)

Q1. Solve the biquadratic equation $x^4 - 3x^2 - 42x - 40 = 0$ by using Ferrari's method.

Q.2 Express $f(x) = \pi x - x^2$ as a half – range sine series in the interval $(0, \pi)$ up to first three terms.

Q.3 Find the inverse Fourier transform of $f(p) = e^{-|p|y}$.

Q.4 Solve the following equation by Z- transform:

$y_{x+2} - 2\cos\alpha\ y_{x+1} + y_x = 0$. $y_0 = 0$. $y_1 = 1$.

Q.5 Find p such that the function f(z) expressed in the polar coordinates as

$f(z) = r^2 \cos 2\theta + i\, r^2 \sin p\theta$ is analytic.

Section - C (5 × 4 = 20)

Q.1 Find the values of c_1 and c_2 such that the function

$f(z) = x^2 + c_1 y^2 - 2xy + i\,(c_2 x^2 - y^2 + 2xy)$ is analytic.

Q.2 Show that the function $u = e^{-2xy}\sin(x^2 - y^2)$ is harmonic. Find the conjugate function v and express u +i v as an analytic function of z.

Q.3 Determine poles and residue at each pole and hence evaluate the integral $\int_c \frac{3z^2 + z}{z^2 - 1} dz$, where c is the circle $|z| = 2$.

Q.4 Find the Laurent's series which represent the function $f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$ when

(a) $2 < |z| < 3$. and (b) $|z| > 3$

Q.5 By residue theorem, evaluate:

$$\int_0^{2\pi} \frac{\cos 2\theta}{5 + 4\cos\theta} d\theta$$

Subject: - Mathematics **Uni. Roll No:-**
Time:-2HOURS 30 Minutes **Total Marks:-80**
Notes:-

1) Attempt **ALL** questions of Section- A, **ANY FOUR** from Section- B ,and **ANY FOUR** from Section- C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

Section:-A $(10 \times 2 = 20)$

Q .1 Cauchy – Riemann equations in polar coordinates .....

Q.2 Value of integral $\int_c \frac{e^{-z}}{z+1} dz$ Where C is the circle $|z| = 1/2$ .....

Q.3 If $f(z)$ have a singular point at $z = a$ then it can be expanded by Taylor's Series or by Laurent's Series .....

Q.4 $f(z) = \sin\frac{1}{1-z}$ at $z = 1$ have the singularity of the type.....

Q.5 Residue of $f(z) = \frac{z^3}{z^2-1}$ at $z=\infty$ is.....

Q.6 Order of each poles of the function $f(z) = \frac{z^3}{(z-1)^4(z-2)(z-3)}$ .....

Q.7 Write the formula for fourier sine transform.....

Q.8 If $f(p)$ is the complex Fourier transform of $F(x)$ then $F[F(ax)] =$.....

Q.9 Z-Transform of unit impulse function is .....

Q-10 If $Z\{f(k)\} = F(z)$, Then $Z\{k^n f(k)\}$ is .....

Section:-B ($4 \times 5 = 20$)

Q1. Prove that $u = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic.

Q.2 Solve the biquadratic equation $x^4 - 3x^2 - 42x - 40 = 0$ by Ferrari's method.

Q.3 Find the Fourier transform of $F(x) = e^{-|x|}$

Q.4 Solve by Z-transform $y_{k+1} + y_k = 1$ if $y_0 = 0$

Q.5 Find the Fourier series of $f(x) = x^2$, $-\pi \leq x \leq \pi$.

Secion.- C ($4 \times 10 = 40$)

Q1. Prove that the function F(z) defined by

$$F(z) = [x^3(1+i) - y^3(1-i)] / [x^2 + y^2] \quad ; \; z \neq 0 \text{ and } F(0) = 0$$

is not regular at the origin although Cauchy – Riemann equations are satisfied there.

Q .2 If $f(z) = u + iv$ is an analytic function of $z = x + iy$ and

$u - v = (x - y)(x^2 + 4xy + y^2)$ then construct f(z) in terms of z .

Q .3 Determine the poles of the function $f(z) = \frac{z-1}{(z+1)^2(z-2)}$ and hence evaluate

$\int_c f(z)\, dz$ by Cauchy's residue theorem where C is the circle $|z - i| = 2$

Q .4 Obtain the Taylor's series expansion of $f(z) = \frac{1}{z^2 - 4z + 3}$ about the point $z = 4$.

Find its region of convergence.

Q .5 Solve the integral $\int_0^{2\pi} \frac{d\theta}{5 - 3\cos\theta}$ using contour integration.